



Protein screening platform speeds rare-earth separation research

Developed at Lawrence Livermore National Laboratory, the SpyCI-LAMBS platform enables scientists to analyse hundreds of lanmodulin protein variants simultaneously without



LLNL scientist Patrick Diep works with a robotic arm to automate the new protein screening method (Credit: www.llnl.gov)

time-consuming purification steps. Researchers have developed a high-throughput protein screening platform that dramatically accelerates research on rare-earth element separation, potentially improv-

ing supply chains for magnets, batteries, and electronic devices. The system reduced testing time for 600 proteins from several years to about one month. Research conducted in California identified protein clusters with improved selectivity for rare-earth metals, and the resulting datasets could support machine-learning models for designing next-generation biomining and metal-recovery technologies.

Battery-free textile powers health sensors

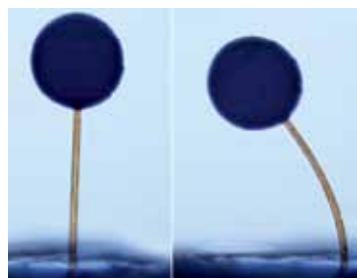
Researchers from the National University of Singapore, in collaboration with the University of Arizona and Tsinghua University, have developed a battery-free smart textile that wirelessly powers wearable health sensors through everyday clothing, enabling continuous blood pressure monitoring without bulky batteries. The system uses ultra-thin epidermal sensors connected through a meta-material fabric that transfers energy from a smartphone while simultaneously transmitting physiological data. The textile maintains accurate sensing even during physical activity by separating power transfer and data communication channels.

Coloured solar films boost efficiency

Researchers at the Fraunhofer Institute for Solar Energy Systems ISE have developed coloured solar films that allow photovoltaic modules to mimic roof tiles and masonry while retaining nearly 95% of the power output of standard panels. The technology combines MorphoColor and Shade-Cut photonic structures to create colour without heavily blocking sunlight, overcoming a major limitation of earlier coloured solar systems. Developed in Freiburg, Germany, the films use microscopic surface patterns inspired by the wings of the Morpho butterfly to selectively reflect visible colours while transmitting most of the solar energy.

Soft magnetic material could help make mini robots and medical devices

Researchers have developed a 3D-printed soft magnetic material capable of bending, gripping, and changing shape in response to magnetic fields, opening possibilities for



The team fabricated ball-and-stick structures. The structures were less than a millimetre in height, with balls that were smaller than a grain of sand. The balls were infused with magnetic particles (Photo Credit: Courtesy of the researchers)

miniature robots and medical devices. Engineered by a team at Massachusetts Institute of Technology (MIT) in collaboration with EPFL and the University of Cincinnati, the technology could support drug delivery, fluid control, and minimally invasive medical tools. The material can form sub-millimetre structures that respond

differently to the same external magnet, enabling precise movement and control without onboard electronics. The researchers fabricated microscopic 'ball-and-stick' structures and bistable magnetic switches to demonstrate the material's multifunctional capabilities.

AI optimises battery cell production efficiency

Research conducted through collaborations involving industrial and academic teams from BMW Group and University of Zagreb, including projects in Croatia and broader



BMW Group and University of Zagreb: Advances in battery cell production using artificial intelligence (Credit: www.press.bmwgroup.com)

lithium-ion manufacturing studies, shows AI can reduce testing cycles by more than half while improving yield consistency. AI-driven software is improving battery cell manufacturing by reducing testing time, cutting material waste, and boost-

ing production efficiency. The technology uses machine learning models to analyse real-time and historical production data, enabling manufacturers to predict battery performance and detect process issues earlier. The approach also supports digital twins and predictive quality control, helping battery makers scale production faster as demand for electric vehicles and energy storage rises globally.